

ZHUOQING ZHONG

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EDUCATION

Carnegie Mellon University

M.S. in Electrical and Computer Engineering
Spring 2027 entering class

Wenzhou-Kean University

B.Sc. in Computer Science

Aug 2022 - Jun 2026

- Overall GPA: 3.93/4.00; Minor: Mathematical Sciences
- Honors: Zhejiang Province Government Scholarship; Dean's First-class Scholarship; Dean's Scholarship for Research and Innovation.

PUBLICATIONS

- Yifan Shen, Chuanmiao Dong, **Zhuoqing Zhong**, et al. "Position: Multimodal LLMs Should Learn from Children", CVPR Workshops (CV4CHL), 2026, pp. 5619-5630.
- Bowen Li*, **Zhuoqing Zhong***, et al. "Improving Negative Rejection Ability in Language Models: A Review of Fine-Tuned LLMs, RAG, and RAFT", Journal of King Saud University Computer and Information Sciences, 38, 21, 2026.
- Ebenezer Fiifi Emire Atta Mills*, Zihui Deng*, **Zhuoqing Zhong***, Jinger Li. "Data-driven prediction of soccer outcomes using enhanced machine and deep learning techniques", Journal of Big Data, 2024.
- Baha Ihnaini, Wei Liu, **Zhuoqing Zhong** (co-first author), et al. "Evaluating Mild Cognitive Impairment Using Large Language Models: A Comparative Analysis", Submitted to Acta Pharmacologica Sinica.

* Equal contribution / shared first authorship.

PROFESSIONAL EXPERIENCE

Institute of Software, Chinese Academy of Sciences

Research Intern

Aug 2024 - Sep 2024

- Implemented a data pipeline and token-level explanations for language models using Layer-wise Relevance Propagation (LRP) and DeepLIFT.
- Compared attribution methods on general and medical text examples through token-relevance visualizations.

RESEARCH EXPERIENCE

Developmental Learning for Multimodal Models

Nov 2025 - Mar 2026

- Surveyed developmental learning for multimodal models, focusing on grounding, supervision inference, memory, and continual adaptation.
- Contributed to a framework spanning task formation, supervision inference, and continuous growth; analyzed limitations in sequential multimodal data and evaluation benchmarks.

LLM Reliability and Rejection Behavior

Nov 2024 - May 2025

Co-first author | Advisor: Dr. Baha Ihnaini

- Conducted a literature review across five scholarly databases, applying inclusion criteria, citation tracing, and structured extraction of models, datasets, and metrics.
- Compared fine-tuning, RAG, and RAFT for rejection behavior, hallucination mitigation, computational cost, and domain adaptability; co-developed a four-dimensional review framework.

Language Models for Cognitive Screening

Feb 2024 - Jul 2025

Co-first author | Advisor: Dr. Baha Ihnaini

- Built a speech-to-text pipeline for cognitive screening, including audio normalization, transcript correction, and a curated Chinese-language subset.
- Compared prompting with supervised fine-tuning and LoRA on Llama-3, Gemma, and Qwen, including unquantized, 8-bit, and 4-bit configurations.
- Implemented balanced 5-fold cross-validation to evaluate screening performance across prompting, adaptation, and quantization settings.

RepoFusion-in-Decoder

Apr 2025 - Aug 2025

- Implemented lightweight encoder fusion and cross-attention to incorporate retrieved repository context into a frozen code language model for cross-file completion.
- Built BM25 retrieval and sliding-window chunking; cached encoder states to reuse repository context across completion requests.
- Benchmarked completion quality and prefill latency on Long Code Arena and RepoEval-Updated; ablated cross-attention depth to assess the quality-efficiency trade-off.

CT Image Reconstruction

Feb 2025 - Jun 2025

- Implemented a U-Net with kernel basis attention and multi-axis fusion for low-dose and sparse-view CT; evaluated MSE, SSIM, and a CNN observer task.

Reinforcement Learning for UAV Planning

Sep 2024 - Oct 2025

- Implemented enhanced Noisy DQN for communication-constrained UAV planning; evaluated against DQN, NoisyNet, and Double DQN in a 15 x 15 grid.

INTERESTS & SKILLS

- Methods: LLM fine-tuning, LoRA, quantization, retrieval-augmented modeling, token attribution.
- Research Interests: Multimodal Large Language Models; Computer Vision; Reinforcement Learning; Reliable Large Language Models.